P0: Compact Kazakh–Russian Translation with Public Parallel Data and NLLB-Generated Supervision

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Working draft for supervisor discussion
31 October 2026

Abstract

We describe P0, a 202-million-parameter autoregressive Transformer for Kazakh-to-Russian translation, initialized from scratch and subsequently continued with synthetic supervision from NLLB-200 3.3B. The locally available base corpus contains 6 851 060 parallel pairs. A source-side monolingual expansion produced 4 000 000 generated records, of which 3 999 605 remain in the filtered archive. The history comprises an initial public-data stage reported to plateau near 17 BLEU and a continuation stage; server training curves and the exact proportions of the base/synthetic mixture are still being recovered. A fresh evaluation on 1 012 FLORES+ devtest sentences yields 20.14 BLEU and 46.48 chrF++. Rescoring the archived outputs of the main competitor, KazRush KK–RU, under the same metric configuration gives 18.89 BLEU and 45.48 chrF++. We document the architecture, data, evaluation, software tests and ONNX export. Results support the usefulness of the system; they do not isolate the causal effect of synthetic data or establish an architecture-only advantage.

1 Introduction

Kazakh-to-Russian translation is a practical setting in which a specialized model can offer local execution and a smaller footprint than a general-purpose language model. Challenges include assembling noisy parallel resources, choosing a compact architecture and evaluating quality reproducibly. The main external reference is `deepvk/kazRush-kk-ru`, a T5-based model trained from random initialization on public parallel data [7, 14]. Its model card reports 197M parameters and 18.8 BLEU on FLORES+ devtest. P0 targets a similar parameter scale with a custom encoder–decoder.

This report provides an implementation-level description, a fresh FLORES+ evaluation, a common rescoring of available benchmarks, and an evidence package containing signatures, hashes, corpus checks, software-test results and reproduction code. The diffusion experiment is outside this draft's scope.

Evidence status. Architecture and defaults are verified against the current repository. Corpus sizes, exact overlaps and primary P0 scores are measured from local artifacts. The plateau near 17 BLEU is an author-reported observation. Server events, original per-stage configurations and the merged-corpus manifest were unavailable. We distinguish these sources rather than reconstructing unobserved training curves.

2 Background and related work

P0 follows the Transformer encoder–decoder [18]; KazRush is a task-specific T5 baseline [7, 14]. P0 combines RMSNorm [19], rotary positional embeddings (RoPE) [16], query/key normalization [3], SwiGLU [15], and decoder grouped-query attention (GQA) [1]. These are established components, not novel attention or normalization methods introduced by this work.

NLLB is the multilingual teacher [8]. Generating targets for source-language monolingual text is *forward translation*, related to sequence-level knowledge distillation [5]. It is not back-translation, which starts from target-language text and synthesizes the source. Tokenization uses SentencePiece [6]; lexical evaluation uses SacreBLEU [13].

3 Data

3.1 Public parallel corpus

Acquisition scripts support OPUS, WMT19 Kazakh–Russian crawl, KazParC and TIL [4, 10, 17]. The filtered training file contains exactly **6 851 060 rows**; a full scan found no invalid TSV column counts. This is the resulting file size, not per-source contributions. KazParC acquisition includes both `kazparc_raw` and `sync_raw`, so public parallel data is not synonymous with entirely human-translated data.

Base filtering removes markup, normalizes to NFC, collapses whitespace, drops empty/identical pairs and one-word sides, applies a 256-whitespace-word ceiling, and deduplicates normalized pairs. Optional stages use fastText language identification and LaBSE alignment scoring [2], with default threshold 0.55. Because stages can be skipped, code alone does not establish the historical filtering settings.

3.2 Synthetic expansion

The monolingual manifest identifies `kz-transformers/multidomain-kazakh-dataset`, revision `7a1fcdf9830b1c34b44b3038aafb672447f41890`, with Leipzig, CC100, OSCAR, news and books. Sentence splitting/filtering excludes strings indexed from the base corpus and both FLORES splits. The manifest records 3 260 296 input documents and 4 000 000 written sentences.

The teacher is `facebook/nllb-200-3.3B`, using `kaz_Cyrl` and `rus_Cyrl`. Generation is sharded with resumable indexed output and validation of missing, duplicate and failed records. The dataset card records generation on six RTX 4090 GPUs. Scanning all six compressed shards confirms **3 999 605 pairs**; metadata records 395 filtered records, retaining successful, nonempty, nonidentical translations.

The repository supports merging base/generated pairs with normalized deduplication and continuing with the existing tokenizer. The final merged file and deduplication totals are not local. We do not assert its exact size or treat the default mixture as verified historical execution. Underlying corpus terms and NLLB's CC-BY-NC-4.0 model license must be considered before redistribution or commercial-use claims.

Table 1: Local data artifacts; synthetic archive size is not merged-corpus size.

Artifact	Rows / sentences
Filtered public parallel training corpus	6 851 060
Selected monolingual Kazakh sentences	4 000 000
Filtered NLLB-generated archive	3 999 605
FLORES+ dev	997
FLORES+ devtest	1 012

3.3 Overlap audit and tokenization

Both complete corpora were checked against FLORES+ dev/devtest after NFC normalization and stripping: **zero exact source matches and zero exact source–reference pair matches** for both corpora/splits. This excludes neither near duplicates nor prior benchmark exposure by teachers/external models, and does not replace auditing the final server-side merged artifact.

The saved shared tokenizer has 32 000 unigram pieces, `nmt_nfkc` normalization, character coverage 0.9995 and a five-million input-sentence sampling limit. IDs are UNK=0, BOS=1, EOS=2, PAD=3. Sources receive BOS, targets BOS/EOS. Both are truncated to 256 subword tokens, forcing terminal target EOS. Teacher forcing shifts the target by one; padding labels are ignored. Length-sorted batches are shuffled. Maximum dev/devtest source lengths are 75/93 tokens; no source is truncated.

4 P0 architecture

P0 has 12 encoder and 12 decoder blocks, hidden dimension 512, head size 64 and FFN hidden size 4096. Direct counting gives **202 016 256 trainable parameters**, excluding rotary buffers. Source embeddings, target embeddings and vocabulary projection share a $32\,000 \times 512$ matrix (Figure 1).

Table 2: Architecture and parameter allocation.

Setting / component	Value
Encoder / decoder layers	12 / 12
Model / head / FFN dimension	512 / 64 / 4096
Encoder self-attention query / KV heads	8 / 8
Decoder self-attention query / KV heads	8 / 4
Cross-attention query / KV heads	8 / 8
Vocabulary / sequence limit	32 000 / 256
Residual dropout	0.1
Shared embedding parameters	16 384 000
Encoder parameters	88 094 208
Decoder parameters	97 537 536
Final decoder RMSNorm parameters	512
Total	**202 016 256**

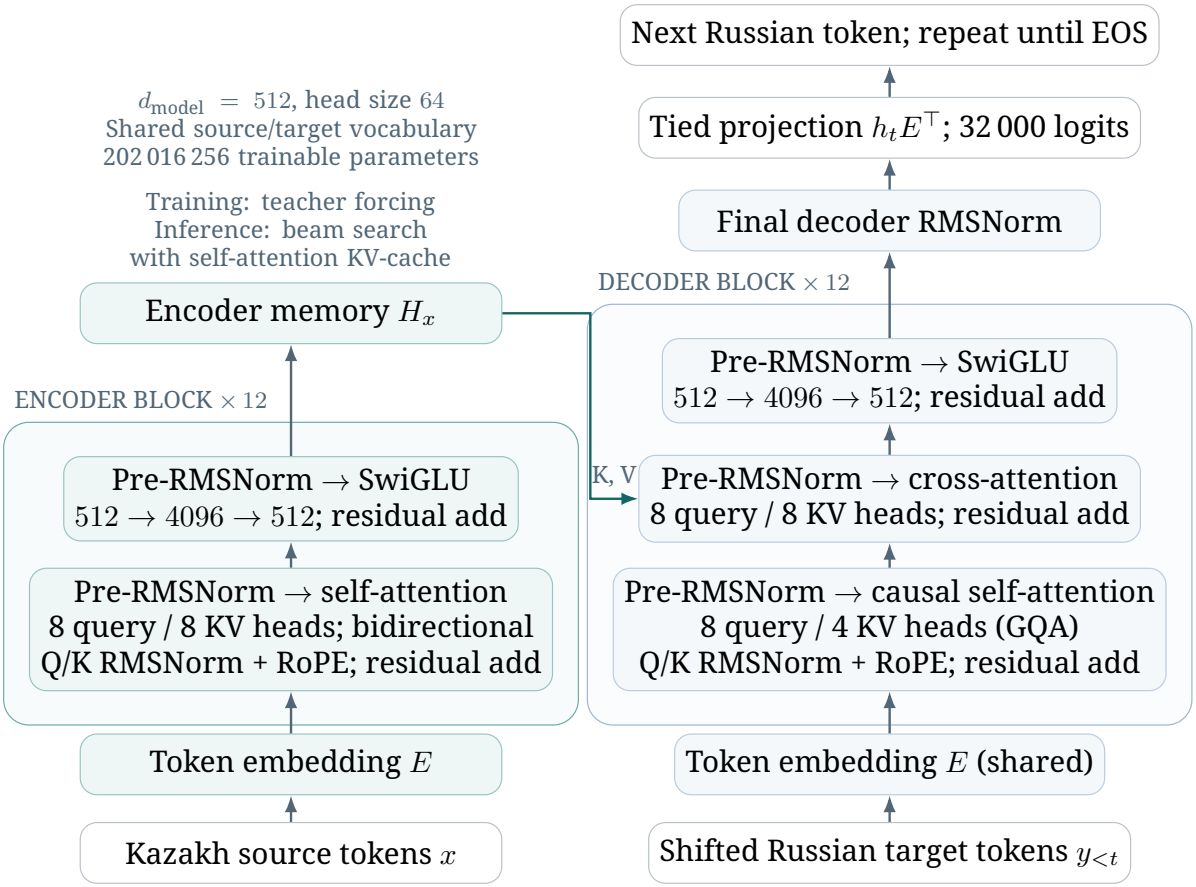


Figure 1: P0, redrawn from the draw.io reference and checked against code. Each pre-normalized attention/FFN sublayer has dropout and a residual add. Block patterns repeat 12 times. There is no extra final encoder norm; the decoder has one. RoPE and Q/K normalization apply to self-attention, not cross-attention. A matching editable draw.io diagram accompanies the source.

4.1 Encoder

Encoder self-attention is bidirectional, with per-head Q/K RMSNorm and RoPE. Each block computes

$$u = h + \text{Dropout}(\text{SelfAttn}(\text{RMSNorm}(h))), \tag{1}$$

$$h' = u + \text{Dropout}(\text{SwiGLU}(\text{RMSNorm}(u))), \tag{2}$$

$$\text{SwiGLU}(z) = (\text{SiLU}(zW_g) \odot zW_u)W_d. \tag{3}$$

4.2 Decoder

Each block contains causal GQA self-attention, cross-attention and SwiGLU, each pre-normalized with a residual connection. Self-attention has eight query and four KV heads; cross-attention has eight of each, without RoPE or Q/K normalization. After final RMSNorm,

$$p_\theta(y_t \mid y_{<t}, x) = \text{softmax}(h_t E^\top). \tag{4}$$

Attention/FFN projections are bias-free. Weights are randomly initialized, not loaded from NLLB or KazRush.

5 Training procedure

5.1 Objective and optimizer

The objective is teacher-forced label-smoothed cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{valid}}} \sum_{(x,y)} \sum_{t \in \text{valid}} \sum_{v=1}^{V} q_\epsilon(v; y_t) \log p_\theta(v \mid y_{<t}, x), \tag{5}$$

$$q_\epsilon(v; y_t) = (1-\epsilon)\mathbf{1}[v = y_t] + \epsilon/V. \tag{6}$$

The current configuration sets $\epsilon = 0.1$. AdamW uses $\beta = (0.9, 0.95)$, weight decay 0.01 for parameters with at least two dimensions, no decay on vectors, and global gradient-norm clipping at 1.0. Accelerate provides mixed precision, accumulation and distributed execution. BF16 describes training autocast. The available checkpoint stores FP32 weights and the fresh evaluation uses FP32; training precision does not imply checkpoint storage precision.

Table 3: Current training defaults verified in code/configuration. Historical per-stage overrides require server logs.

Setting	Current value
Peak / minimum learning rate	2×10^{-4} / 5×10^{-5}
Warmup / schedule	2 000 optimizer steps / cosine
Micro-batch / accumulation	16 pairs per GPU / 4
Effective batch for six GPUs	384 pairs
Precision / seed	BF16 autocast / 42
Configured epochs	3
Evaluation / checkpoint interval	2 000 optimizer steps

5.2 Two-stage history

1. **Public-data stage.** P0 was trained from scratch on public parallel data. The author reports a plateau near **17 BLEU**. Its exact split, step, score series and stopping criterion await the logs.
2. **Synthetic-data continuation.** Approximately four million Kazakh sentences were translated with NLLB-200 3.3B, filtered and used for continued P0 training on a mixture of original and synthetic pairs, as confirmed by the author. The documented workflow restores model/optimizer, resets the data cursor and creates a fresh schedule. The exact mixture proportions, sampling weights, epochs and duration remain to be confirmed.

The available best export records global step 116 000, which alone cannot establish per-stage steps/epochs. Its stored 20.3053 BLEU lacks an explicit split identifier. A fresh dev run reproduces all four stored metrics exactly, corroborating a dev origin; fresh devtest is reported separately below. This history motivates a synthetic-supervision hypothesis but is not a controlled ablation. The reported plateau and final checkpoint cannot be treated as a precisely measured before/after gain under a verified common protocol; the pre-continuation checkpoint must be reevaluated.

5.3 Selection and observability

The current trainer selects the best dev export by spBLEU when available, otherwise chrF++. Full step checkpoints retain optimizer, scheduler and data cursor; the best directory is a weight export. TensorBoard/JSONL log loss, learning rate, gradient norm, step time, throughput and GPU memory. Training curves are absent because server events are unavailable. Recovering them will allow real convergence and compute-cost reporting. Software checks in Appendix C test implementation behavior separately from translation quality.

6 Evaluation protocol

Local Kazakh/Russian Cyrillic FLORES+ files [9] have 997 dev and 1 012 devtest pairs. Dev is the intended selection split; devtest is the reporting split. SHA-256 fixes local file identities; the original upstream revision remains to be recovered.

Fresh P0 evaluation uses `checkpoints/p0/best/model.pt`, beam 5, source/generation limits 256, batch 16, and the repository decoder's length-normalization exponent 0.6. Self-attention uses a KV-cache. Settings were fixed before this full devtest run; there was no devtest decoding sweep. Hardware is Apple M4 Pro, 24 GiB unified memory, MPS, PyTorch 2.14.0 and FP32 parameters.

6.1 Metrics

We compute case-sensitive BLEU with `13a` tokenization, chrF, chrF++, and TER using SacreBLEU 2.6.0 [11–13], on 0–100 scales. chrF++ adds word bigrams to character n-grams; it differs from chrF. Lower TER is better. Full signatures are in Appendix D. No extra postprocessing is added beyond translation decoding/stripping and metric-specific normalization. COMET/spBLEU are not newly measured here.

6.2 Artifacts and confidence

KazRush is the main task-specific comparator. Its saved hypothesis file has 1 012 nonempty lines, but no source IDs or original generation manifest. Positional alignment with FLORES is assumed; exact decoding settings cannot be reverified. Rescored 18.89 BLEU / 48.70 chrF is consistent with the card's 18.8 BLEU / 48.7 chrF, although this is not independent alignment proof.

Other saved runs provide context. Indexed JSONL must match source/reference at every index; incomplete archives are excluded. No paid API requests are made. Identifiers are reported as saved; versions, hardware and original inference settings are not controlled across systems. Paired bootstrap uses 1 000 resamples, seed 12 345, P0 versus KazRush. It is conditional on archived alignment and reflects test-set sampling uncertainty, not training-seed variation.

7 Results

7.1 Fresh evaluation on both splits

Table 4: Same P0 checkpoint and fixed decoding settings. Dev is not a second held-out test. The dev row exactly reproduces the checkpoint's stored metrics.

Split	Sentences	BLEU ↑	chrF ↑	chrF++ ↑	TER ↓
Dev	997	20.31	49.46	46.40	69.72
Devtest	1 012	20.14	49.55	46.48	69.84

7.2 Primary comparison

Table 5: FLORES+ devtest: 1 012 sentences, common scoring. P0 is newly generated; KazRush is rescored from a positional archive.

System	Params	BLEU ↑	chrF ↑	chrF++ ↑	TER ↓
KazRush KK–RU	197M[†]	18.89	48.70	45.48	71.47
P0	202.0M	**20.14**	**49.55**	**46.48**	**69.84**

[†]Count from the KazRush model card.

P0 improves over the archive by **1.25 BLEU** and **1.00 chrF++**, with TER lower by 1.62 points. Systems differ in architecture, tokenization, data and schedule; P0 has additional NLLB supervision. This is a system-level comparison, not an architecture-only ablation.

Table 6: Paired bootstrap: bootstrap mean and approximate 95% interval half-width returned by SacreBLEU. p compares against KazRush.

Metric	System	Score	Bootstrap mean ± half-width	p
BLEU	KazRush	18.89	18.88 ± 0.96	—
BLEU	P0	20.14	20.12 ± 0.96	0.0010
chrF2++	KazRush	45.48	45.47 ± 0.73	—
chrF2++	P0	46.48	46.47 ± 0.74	0.0010

Both metrics have paired $p = 0.0010$ under the alignment assumption. Marginal intervals can overlap despite the paired result, because the paired test resamples the same sentences for both systems. It does not address checkpoint selection, seed variance or hidden test exposure.

7.3 Context and examples

Appendix A inventories all complete, aligned saved runs. Some hosted/larger systems score above P0, including Google Translate and the batched `gpt-6-luna` archive. The low saved `Qwen3-0.6B` score describes that prompted run, not the best attainable family performance. This collection is not a state-of-the-art claim.

Appendix B shows fixed devtest rows 1 and 3, without ranking by sentence scores. Row 1 expresses the main medical claim with wording different from the reference. In row 3, P0 produces *эти ярости* where the reference has *эти открытия*, a lexical error within plausible Russian. Corpus gains do not eliminate meaning-changing errors, and lexical overlap is not a full adequacy assessment.

8 Deployment and verification

8.1 ONNX

Separate FP32 encoder and full-prefix decoder ONNX graphs use opset 18, dynamic batch/sequence axes, and companion tokenizer metadata. The export decomposes RMSNorm/attention into portable operations. Encoder output is reused across decoder calls. The current export *does not expose a decoder KV-cache*: each step resubmits the full prefix. The FP32 bundle is about 898 MiB, with shared embeddings independently represented in the separate graphs and an explicit decoder output projection. This is functional portability, not an optimized Android release.

Three short Kazakh inputs are run end to end through ONNX Runtime CPU, comparing every greedy next-token decision against native PyTorch. All three sequences agree; details are in the evidence package. This is not a full-corpus export-quality evaluation. Android latency/RAM, FP16/INT8 quality and accelerator support are unmeasured.

8.2 Local inference

The full devtest generation takes 128.0 seconds in this MPS session, or 7.91 sentences/s at batch 16, including the first batch and excluding model loading/metrics. This is a single throughput observation, not warmed single-request latency or an Android estimate.

9 Limitations and next experiments

Limitations include missing server provenance, one final checkpoint, one inference platform and the positional KazRush archive. The plateau is not tied to a recovered evaluation manifest. Actual mixture size, historical overrides, durations and GPU-hours are unknown. Exact overlap checks do not exclude near duplicates or teacher exposure. No human adequacy/fluency study has been performed. The next draft should:

1. recover event files, JSONL, per-stage configurations, run manifests and merged-corpus statistics;
2. evaluate the pre-NLLB checkpoint with identical files and decoding;
3. regenerate KazRush with a pinned revision, explicit settings and IDs;
4. compare public-only and mixed-data continuation at matched token/update budgets, ideally across several seeds;
5. tune on dev, add documented neural/human evaluation and confirm on independent unseen material;
6. export a cache-aware decoder and measure mobile quality, memory, latency and energy on a named Android device.

Repeated development on observed devtest weakens its held-out role; major future choices should use dev and independent confirmation.

10 Conclusion

P0 is a 202.0M-parameter translator with a public-data and NLLB-supervision training history. Fresh devtest evaluation gives 20.14 BLEU versus 18.89 for the rescored KazRush archive. The evidence supports a useful compact system. Logs and matched continuation experiments are needed to quantify the synthetic-data contribution.

References

[1] Joshua Ainslie et al. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *EMNLP*, 2023. URL https://arxiv.org/abs/2305.13245.

[2] Fangxia Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic BERT sentence embedding. *arXiv preprint arXiv:2007.01852*, 2020. URL https://arxiv.org/abs/2007.01852.

[3] Alex Henry, Prudhvi Raj Dachapally, Shubham Shantaram Pawar, and Yuxuan Chen. Query-key normalization for transformers. In *Findings of EMNLP*, 2020. URL https://aclanthology.org/2020.findings-emnlp.379/.

[4] Institute of Smart Systems and Artificial Intelligence. KazParC: Kazakh parallel corpus. Hugging Face dataset, n.d. URL https://huggingface.co/datasets/issai/kazparc.

[5] Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *EMNLP*, 2016. URL https://aclanthology.org/D16-1139/.

[6] Taku Kudo and John Richardson. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *EMNLP: System Demonstrations*, 2018. URL https://aclanthology.org/D18-2012/.

[7] Anna Lebedeva and Andrey Sokolov. kazRush-kk-ru: Translation model from kazakh to russian. Hugging Face model card, 2024. URL https://huggingface.co/deepvk/kazRush-kk-ru. Accessed 31 October 2026; snapshot included in the evidence package.

[8] NLLB Team et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022. URL https://arxiv.org/abs/2207.04672.

[9] Open Language Data Initiative. FLORES+: A multilingual machine translation evaluation dataset. Dataset repository, 2026. URL https://huggingface.co/datasets/openlanguagedata/flores_plus. Local dev/devtest snapshots identified by SHA-256 in this report.

[10] OPUS Project. OPUS: Open parallel corpus. Corpus collection, n.d. URL https://opus.nlpl.eu/. Source collection used by the project's acquisition pipeline.

[11] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: A method for automatic evaluation of machine translation. In *ACL*, 2002. URL https://aclanthology.org/P02-1040/.

[12] Maja Popović. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, 2015. URL https://aclanthology.org/W15-3049/.

[13] Matt Post. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation*, 2018. URL https://aclanthology.org/W18-6319/.

[14] Colin Raffel et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL https://jmlr.org/papers/v21/20-074.html.

[15] Noam Shazeer. GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020. URL https://arxiv.org/abs/2002.05202.

[16] Jianlin Su et al. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021. URL https://arxiv.org/abs/2104.09864.

[17] Turkic Interlingua. TIL-MT: Turkic parallel corpus. Dataset repository, n.d. URL https://github.com/turkic-interlingua/til-mt.

[18] Ashish Vaswani et al. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. URL https://arxiv.org/abs/1706.03762.

[19] Biao Zhang and Rico Sennrich. Root mean square layer normalization. *Advances in Neural Information Processing Systems*, 2019. URL https://arxiv.org/abs/1910.07467.

A Complete local benchmark inventory

Table 7: Common rescoring of complete devtest artifacts. API/model names are archival identifiers, not verified backend identities. Only P0 is rerun.

System / identifier	BLEU ↑	chrF ↑	chrF++ ↑	TER ↓
P0 (fresh evaluation)	20.14	49.55	46.48	69.84
KazRush KK-RU	18.89	48.70	45.48	71.47
Google Translate	21.83	49.49	46.62	67.02
Qwen3-0.6B	0.99	16.39	13.89	100.71
gpt-6-luna (batched)	24.33	53.71	50.57	65.16
qwen/qwen3.8-27b	21.87	51.24	48.10	69.66
google/gemma-4-26b-a4b-it	21.69	51.39	48.21	69.23
google/gemma-4-31b-it	18.84	50.32	47.10	94.65

Every row has 1 012 hypotheses. Indexed runs have source/reference checks. KazRush and Google Translate rely on positional alignment. Service costs, batching and latency are not comparable with P0. The incomplete single-request `gpt-6-luna` archive is excluded, as is the unidentified legacy `reports/devtest.hyp.txt` (1 011 lines). Error-only API experiments are not benchmark results.

For external context, the KazRush card reports NLLB-200 3.3B at 21.5 BLEU / 50.7 chrF, and KazRush at 18.8 BLEU / 48.7 chrF. These are published numbers, not newly rescored rows: no complete local NLLB devtest output was found. Model-card COMET values are not combined with unmeasured P0 COMET.

B Qualitative examples

Text is retained from saved outputs; row numbers are one-based.

Example 1 (devtest row 1). Source. «Қазір бізде диабетпен ауырған, бірақ қазір диабеті жоқ 4 айлық тышқандар бар», - деп қосты ол.

Reference. "Теперь у нас есть четырёхмесячные мыши, у которых больше нет диабета", — добавил он.

P0. "Сейчас у нас есть 4-месячные мыши, которые болели диабетом, но сейчас не имеют диабета", - добавил он.

KazRush. «Сейчас у нас есть 4-х месячные мыши, которые болели диабетом, но сейчас нет диабета», - добавил он.

Example 3 (devtest row 3). Source. Кейбір басқа сарапшылар сияқты, ол диабетті емдеуге болатынына күмәнмен қарайды, бұл ашулардың 1-түрге жататын диабетімен ауыратын адамдарға қатысы жоқ екенін ескертті.

Reference. Как и некоторые другие эксперты, он сомневается, что диабет излечим, отмечая, что эти открытия не относятся к людям, у которых уже развился диабет 1 типа.

P0. Как и некоторые другие эксперты, он скептически относится к тому, что диабет можно вылечить, предупреждая, что эти ярости не имеют отношения к людям с диабетом 1 типа.

KazRush. Как и некоторые другие эксперты, он скептически относится к тому, что диабет можно лечить, предупредив, что гнев не связан с людьми с диабетом 1 типа.

C Software-test inventory

All **29 tests pass**. Pytest discovers five function-based checks missed by unittest's 24-test discovery. These are not translation benchmarks. The nine diffusion tests inventory the full repository, without adding a diffusion study to this report.

Table 8: Software verification by area.

Area	Passing tests
Training continuation / scheduler	3
Translation metrics / alignment	4
Synthetic merge / failure preservation	2
NLLB sharding / resume / validation	6
Monolingual filtering	3
Benchmark sampling	2
Diffusion implementation (outside report scope)	9
Total	**29**

Table 9: Individual checks; full output and JUnit XML accompany the source.

Module	Test	Status
test_diffusion	test_adarmsnorm_uses_timestep_conditioning	Pass
test_diffusion	test_base_config_remains_valid_without_diffusion_section	Pass
test_diffusion	test_corruption_masks_at_least_one_token_per_nonempty_row	Pass
test_diffusion	test_corruption_masks_valid_tokens_and_fully_masks_terminal_step	Pass
test_diffusion	test_decoder_is_bidirectional_and_outputs_only_vocabulary_logits	Pass
test_diffusion	test_lengths_include_all_nonpadding_tokens	Pass
test_diffusion	test_masked_loss_only_scores_corrupted_positions	Pass
test_diffusion	test_reverse_step_reveals_high_confidence_then_finishes	Pass
test_diffusion	test_translation_runs_all_reverse_steps_and_returns_text	Pass
test_merge_synthetic	test_failed_merge_preserves_previous_output	Pass
test_merge_synthetic	test_merge_deduplicates_and_is_reproducible	Pass
test_metrics	test_calculates_lexical_metrics	Pass
test_metrics	test_p0_evaluation_includes_ter	Pass
test_metrics	test_records_optional_metric_import_errors	Pass
test_metrics	test_rejects_misaligned_rows	Pass
test_monolingual	test_split_sentences_normalizes_whitespace	Pass
test_monolingual	test_kazakh_filter_rejects_links_and_short_noise	Pass
test_monolingual	test_sentence_hash_is_stable	Pass
test_nllb_pipeline	test_length_bucketing_keeps_indices_and_groups_similar_lengths	Pass
test_nllb_pipeline	test_rank_shards_cover_nonempty_input_without_overlap	Pass
test_nllb_pipeline	test_resume_compacts_duplicate_successes	Pass
test_nllb_pipeline	test_resume_keeps_successes_and_removes_errors_and_partial_record	Pass
test_nllb_pipeline	test_validator_can_reuse_its_sqlite_index	Pass
test_nllb_pipeline	test_validator_rejects_source_index_mismatch	Pass
test_prepare_translation_benchmark	test_bucket_boundaries	Pass
test_prepare_translation_benchmark	test_reservoir_is_deterministic	Pass
test_train_resume	test_exact_resume_keeps_checkpoint_cursor_and_phase_progress	Pass
test_train_resume	test_new_data_phase_resets_cursor_but_keeps_global_step	Pass
test_train_resume	test_new_data_schedule_restarts_without_resetting_optimizer_state	Pass

D Reproduction and provenance

Run from the repository root. Evidence scripts use PyTorch, SentencePiece, PyYAML, SacreBLEU, NumPy and pytest. Export verification also needs ONNX and ONNX Runtime. Versions are saved in `evidence/environment.json`.

```
PYTHONPATH=src .venv/bin/python -m pytest tests -v
PYTHONPATH=src .venv/bin/python docs/p0-report/collect_evidence.py archives
PYTHONPATH=src .venv/bin/python docs/p0-report/collect_evidence.py corpora
PYTHONPATH=src .venv/bin/python docs/p0-report/collect_evidence.py p0 --split dev
PYTHONPATH=src .venv/bin/python docs/p0-report/collect_evidence.py p0 --split devtest
PYTHONPATH=src .venv/bin/python docs/p0-report/summarize_results.py
```

Generation appends indexed records and can resume. For a fresh independent rerun, move the corresponding `reports/p0-report/p0-<split>.jsonl` aside first. The model/evaluation source revision is `b132f12c91dff745db5e9383b0f16eee46b2541c`; report code is committed separately.

Metric signatures.

```
BLEU:
nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.6.0
chrF:
nrefs:1|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.6.0
chrF++:
nrefs:1|case:mixed|eff:yes|nc:6|nw:2|space:no|version:2.6.0
TER:
nrefs:1|case:lc|tok:tercom|norm:no|punct:yes|asian:no|version:2.6.0
```

Primary SHA-256 identities.

```
P0 checkpoint:
e336d8b8401ab2dab35d1fab305323de752ad5fe73250c83ff8b82922cae9bb3
FLORES+ devtest:
867dcb0ac62b096821516eff462d18efd1b049a9c17715026f755d7d00e21bcb
FLORES+ dev:
315f7defe72924dc3c87c1e69107d4efeb54eacbc9b3fe2bcfa8c74a5386d352
KazRush archive:
0a29b50e9372eacccf01b55cc02c694ff1feb7a6ab29fe234a443f2b6ac0ad73
P0 tokenizer:
2edd3aa510364694cd29b4fde437ef54b33c6e70ed1173be71b17de292ea727a
```

Evidence includes config, original checkpoint metadata, downloaded KazRush card/config, synthetic metadata, monolingual manifest, corpus hashes/counts, bootstrap, examples and tests.

Before submission. Recover per-stage training manifests, hardware, GPU-hours, mixtures and the plateau evaluation; pin the upstream evaluation revision; regenerate KazRush. Add curves only from real records. This draft is intended to begin supervisor feedback.